



Sita Ram Prasad Martha

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Professional Summary

- Full Stack Python Developer with AI/ML expertise and Graduate Research Assistant with hands-on experience building real-time, data-driven systems using **Python, FastAPI, and React.js**.
- Experienced in end-to-end machine learning workflows using **PyTorch and OpenCV**, including dataset preparation, model training, evaluation (mAP, precision/recall), and real-time inference optimization using **ONNX Runtime**.
- Developed RESTful APIs and streaming services with **FastAPI** and WebSockets, enabling high-frequency telemetry ingestion and interactive visualization dashboards.
- Built responsive frontend applications using **React, TypeScript, Redux, and Tailwind CSS**, integrating secure backend APIs in cloud-native environments.
- Familiar with **Docker, Azure DevOps, and CI/CD pipelines**, supporting containerized deployments and structured development workflows in Agile teams.
- Microsoft Certified: **Azure Data Fundamentals (DP-900)**, with strong foundational knowledge in cloud data services and modern application architecture.

Technical Skills

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Machine Learning & Computer Vision:** PyTorch, OpenCV, YOLO, Faster R-CNN, ONNX, ONNX Runtime, NumPy, Pandas, Dataset Labeling & Augmentation, mAP/Precision/Recall Evaluation, Inference Optimization
- **Backend Development & APIs:** FastAPI, Flask, RESTful APIs, WebSockets, JSON Schema Design, Real-Time Streaming, Telemetry Ingestion, Authentication (JWT), CRUD Operations
- **Frontend Development:** React.js (Hooks, Redux, Context API), TypeScript, Tailwind CSS, Responsive Design (Flexbox, CSS Grid), Axios, Fetch, Accessibility (ADA/WCAG)
- **Databases:** PostgreSQL, MySQL, SQL Query Optimization, Data Modeling
- **DevOps & Deployment:** Docker, Azure DevOps, Azure App Service, CI/CD Pipelines, Git, GitHub, Azure Repos
- **Build & Tooling:** Webpack, Vite, Jupyter Notebook, Chrome DevTools, React DevTools, Lighthouse
- **Systems & Environment:** Linux, Real-Time Inference Systems, Simulator Integration, Structured Logging, Performance Benchmarking

Education

North Carolina A&T State University, MS in Computer Science

Aug 2023 – Dec 2025

- GPA: 3.79/4.0

Vidya Jyothi Institute of Technology, BS in Computer Science

Jul 2018 – May 2022

- GPA: 7.96/10.0

Professional Experience

NC A&T State University, Greensboro, NC

Jan 2024 – Dec 2025

Graduate Research Assistant – CR2C2 (NCDOT/UTC Funded)

- Architected a full-stack, real-time animal detection platform by integrating a Python-based ML inference service with a high-fidelity rural driving simulator and a web dashboard built using **HTML5**, **CSS3**, **Tailwind CSS**, and **JavaScript** for live visualization of animal-vehicle interactions.
- Led the end-to-end ML workflow using **Python**, **PyTorch**, **OpenCV**, **NumPy**, and **Pandas**; curated and labeled rural-road image datasets from simulator and external sources, applied augmentation, and trained YOLO-style and region-based object detection models optimized for low-light and adverse-weather conditions.
- Built a Python-based data ingestion layer integrating simulator APIs and log streams to capture real-time vehicle telemetry (**speed**, **brake pressure**, **throttle/acceleration**, **steering angle**) alongside detection outputs (class, confidence, **distance to animal**, bounding-box coordinates).
- Exposed simulator and model outputs through **REST** and streaming endpoints using **FastAPI**, defining structured JSON schemas for historical replay and live telemetry/detection subscriptions.
- Implemented a lightweight **WebSocket**-based channel (FastAPI backend + JavaScript client) to stream high-frequency telemetry and detection events with low-latency feedback for interactive driving scenarios.
- Developed a responsive telemetry dashboard using **HTML templates**, **Tailwind CSS**, and modular JavaScript components, featuring live video panels with **OpenCV** detection overlays, speedometer, brake/acceleration time-line, steering-angle gauge, and synchronized distance-to-animal indicators.
- Built client-side JavaScript logic to parse streaming JSON payloads, normalize units, smooth sensor noise, and synchronize frames, telemetry, and detection events for accurate driver-behavior analysis.
- Optimized inference by exporting trained models to **ONNX** and deploying with **ONNX Runtime** on GPU-enabled hardware, tuning input resolution, batch size, and confidence thresholds to meet real-time latency constraints.
- Containerized backend services using **Docker** and managed source control via **Git** and **GitHub**, structuring modular repositories for data pipelines, training scripts, API services, and frontend assets to ensure reproducibility and collaboration.
- Produced architecture diagrams and experiment logs using **Markdown**, **Jupyter Notebooks**, and **CSV/Excel** reports to track model performance (mAP, precision, recall, latency) and communicate design trade-offs to faculty advisors, NCDOT partners, and UTC stakeholders.

Environment: Python, PyTorch, OpenCV, NumPy, Pandas, FastAPI, HTML5, CSS3, Tailwind CSS, JavaScript, WebSockets, ONNX, ONNX Runtime, Docker, Git, GitHub, Jupyter Notebook, CSV/Excel, Linux.

ValueLabs, Hyderabad, India

Mar 2022 – Jul 2023

Software Developer

- Contributed to the end-to-end frontend and backend implementation for the rebranding of a client's public website, supporting both **React.js**-based UI development and **Python (FastAPI/Flask)** backend services in a cloud-native environment.
- Translated **Figma design systems** into responsive, production-ready components, ensuring consistent typography, spacing, and branding across multiple pages.
- Implemented **ADA/WCAG-compliant** accessibility using semantic **HTML5**, **CSS3**, ARIA attributes, and keyboard navigation patterns.
- Managed application state using **Redux** and **React Context API**, supporting shared layout state and multi-step marketing and lead-generation flows.
- Integrated frontend components with backend **RESTful APIs** using **Axios**, **Fetch**, and **async/await**, handling structured loading and error states.
- Developed backend services using **Python** and **FastAPI/Flask**, building REST APIs for form submissions, data retrieval, and business logic processing.
- Designed and optimized database interactions using **PostgreSQL/MySQL**, writing SQL queries and implementing CRUD operations.
- Assisted in implementing authentication and authorization mechanisms using **JWT** and role-based access control.
- Optimized frontend builds using **Webpack** and **Vite**, applying code-splitting, lazy loading, and bundle optimization to improve performance.
- Maintained source control using **Git** and **GitHub/Azure Repos**, following feature branching and pull request workflows in an Agile environment.
- Participated in **CI/CD** processes using **Azure DevOps**, supporting automated build and deployment to **Azure App Service**.

- Debugged cross-browser and performance issues using **Chrome DevTools**, **React DevTools**, and Lighthouse audits.

Environment: React.js, JavaScript (ES6+), TypeScript, Redux, Context API, Python, FastAPI/Flask, PostgreSQL/MySQL, HTML5, CSS3, Tailwind CSS, REST APIs, Axios, Fetch, Webpack, Vite, Azure DevOps, Azure App Service, Git, GitHub, Azure Repos, Chrome DevTools, React DevTools, Lighthouse, CI/CD.

Technical Projects

Learning Management System (Undergrad Project)

- Developed a scalable learning management system using a React frontend and Node.js backend, supporting over 5,000 users concurrently.
- Designed PostgreSQL schemas for user profiles, course content, and progress tracking, ensuring fast queries and reliable data integrity.
- Implemented JWT-based secure login and role-based permissions for students, instructors, and admins.
- Optimized frontend components and API response times, reducing course access latency by 70
- Deployed the platform with Docker on AWS EC2 with CI/CD pipelines in GitHub Actions for streamlined updates.
- **Tools Used:** React.js, Node.js, Express.js, PostgreSQL, JWT, Docker, AWS EC2, GitHub Actions

AI-Driven Internal Coding Analysis Platform

- Built a secure web-based platform enabling developers to write, execute, and debug code snippets in real time with React.js frontend and FastAPI backend.
- Integrated an AI-powered analysis engine using OpenAI GPT-4 to detect bugs, explain root causes, and suggest optimized solutions for submitted code.
- Designed PostgreSQL schemas to store code history, execution logs, and AI recommendations, allowing users to revisit and compare code versions.
- Provided step-by-step explanations of detected issues and generated fully optimized code samples with best practices in Python, JavaScript, and SQL.
- Implemented role-based access with JWT for secure collaboration among developers, team leads, and code reviewers.
- Deployed the platform on AWS ECS with Docker containers, scaling to support simultaneous sessions for over 1,000 active users.
- **Tools Used:** React.js, FastAPI, GPT-4, PostgreSQL, JWT, Docker, AWS ECS, GitHub Actions

Intelligent Knowledge Graph-Driven Query Assistant

- Developed a knowledge assistant leveraging LangChain with RAG architecture, combining Pinecone vector search with GPT-4-based question answering.
- Designed FastAPI services integrating OpenAI endpoints and handling advanced prompt flows with fallback mechanisms.
- Built a Next.js frontend with real-time chat UI and conversation memory for contextual responses.
- Orchestrated model training and data pipelines using Airflow DAGs to refresh vector stores with updated domain content.
- **Tools Used:** LangChain, GPT-4, FastAPI, Pinecone, Next.js, Airflow, Swagger, JWT

Certifications

- **Microsoft Certified: Azure Data Fundamentals (DP-900)** – Microsoft
- **PCAP: Programming Essentials in Python** – Python Institute (OpenEDG)
- **CLA: Programming Essentials in C – C++** Institute (Cisco Networking Academy)
- **Database Programming with PL/SQL** – Oracle Academy
- **Technical Support Fundamentals** – Google (Coursera)
- **Complete HTML5 Course: From Beginner to Expert** – Udemy